Toza va ultra toza suv tizimi qurilmasidan foydalanish bo'yicha qo'llanma



Ishlab chiqaruvchi	Human Corporation
Mahsulot nomi (Cat. No.)	Sof va ultra toza suv tizimi (POWER I)

1. Yo’riqnoma

Pure Power seriyali Pure & Ultra Water System, vodoprovod suvidan toza suv (III turdagi) va ultra toza suv (I turdagi) suv ishlab chiqaradi.

Pure Power RO filtri va Ion filtrini birlashtirgan. U har bir funksional bosqich orqali alohida RO bosqichi va UP (ultra tozalash) bosqichini bajarishi mumkin.

1-1-jadval. Suvni tozalash usuli

Noorganik ionlar	(1) Ion almashinuvi (2) Distillash (3) RO
Organik moddalar	(1) RO (2) Distillash (3) Faol uglerod
Zarrachalar	(1) MF filtri (2) RO (3) Distillash
Mikroorganizmlar	(1) MF filtri (2) RO (3) Distillash (4) UV

1-2-jadval. O'rnatilgan filtrlar

Pre-filtr	Uglerod filtri	RO Membran	HP-Pack	Yakuniy filtr	UV chiroq	UF filtri
1	1	1	1	1	Qo’shimcha	Qo’shimcha

1-3-jadval. Quvvat sig'imi

Model	Sig'imi	Mahsulot sifati
Power I	10 ~ 15 L/Soat	III toifa va I toifa
Power II	25 ~ 30 L/Soat	
Power III	45 ~ 50 L/soat	

Ⅱ. Texnik xususiyatlar

1. Xususiyat.

1) ASTM, CAP, NCCLS standart I turdagi suv uchun mos keladi.

2) Suvdagi ifloslanishni bartaraf etish

3) Reaktiv darajasidagi suv HPLC, AA, IC, ICP va BOD sinovlari kabi eng muhim laboratoriya sinovlari uchun yetarlicha toza.

2. Texnik xususiyatlar

1) Ishlab chiqarish tezligi (Product rate):

15 L/soat (maks.: II seriya), 30 L/soat (maks.: II seriya), 50 L/soat (maks.: III seriya)

2) Suv sifati (Product Water Quality):

Ultra toza suv uchun: 18,3 MΩ·cm (ko‘rsatkich)

Toza suv uchun: 0–25 µS/cm (ko‘rsatkich)

3) Nazoratchi: mikro tip

4) Umumiy organik uglerod: <10 ppb / <5 ppb (Tanlov: UV chiroq)

5) Zarrachalarsiz (@ 0.2㎛): <1/ml

6) Umumiy erigan qattiq moddalar: <20 ppb

7) Silikat: <0.1 ppb

8) Og'ir metallar: < 1 ppb

9) Bakteriyalar: < 1 cfu/ml

10) Endotoksin: < 0,1 eu/ml

11) Pirogenlar: Salbiy

12) ND (Aniqlanmagan):

Ca, Mg, K, Al, Cr, Cu, Pb, Hg, As, Se, Cd, Ag, Ba, Zn, Sb, Be, Tl, Ftorid, Nitrat, Sulfat, Erkin Xlor, Xloraminlar

13) Komplektdagi kartrijlar (Included Cartridge):

Pre-filtr (1 dona)

Aktiv uglerod filtri (1 dona)

R/O membrana filtri (1 dona)

HP-Pack (1 to‘plam) 0,22 µm yakuniy filtr (1 dona)

14) RO membranasi va HP-Pack almashinuvi uchun signalizatsiya

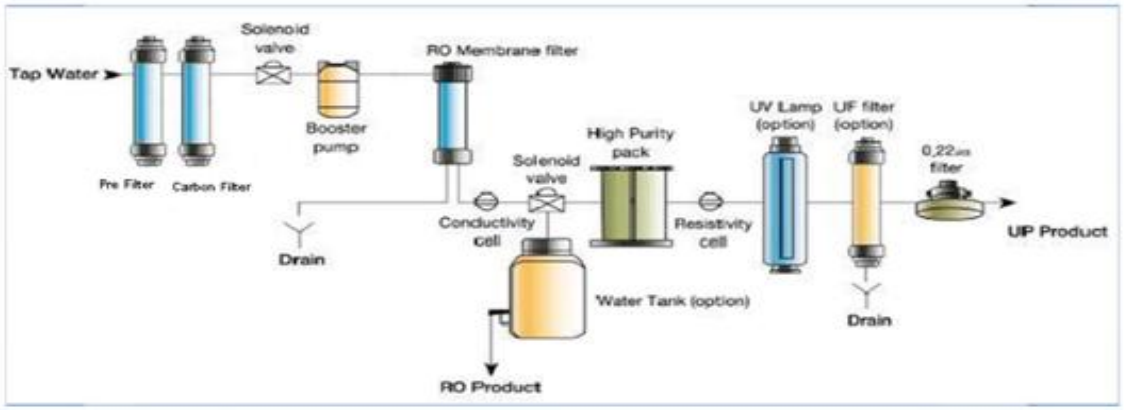
15) Nasos materiali: PVDF diafragma nasosi

16) O’lchamlari 450 (eni) × 400 (chuqurligi) × 580 (balandligi) mm

17) Quvvat: 220Volt, 50/60Hz, bitta fazali

Ⅲ. Diagramma va aksessuarlar

1. Diagramma



2. Aksessuarlar

Yo'q	Qism nomi	Qism raqami
1	Pre - filtr	HM –PF10
2	Uglerod filtri	HM –AC10
3	RO membranasi	HM-200 (I seriyasi) HM-3012 (II, III seriyalar)
4	HP-Pack	HM –HP
5	UV lampa	HM-UV
6	UF filtri	HM-UF
7	Drenaj klapani	HM –DV
8	Kuchaytirgich nasosi	HM –R130 (I, II seriyalar) HM –R300 (II seriya)
9	Solenoid klapan	HM –SV
10	0.22 Yakuniy filtr	HM –022FC
11	3 tomonlama solenoid klapan	HM-SV3
12	Boshqaruv paneli va monitoringi	HM –MB
13	Darajali sensor	HM –LS
14	Quvvat kodi	HM –PC

IV. Filtrni almashtirish 1: Pre filter



1. **Pre Filter** : When pre filter color is yellow(average : once a month or two months)
2. **Carbon Filter** : Carbon filter is exchanged one time when Pre filter is exchanged two times

3. HP-Pack : When the resistivity is below 10㏁-cm (HP-Pack Lamp "ON")

4. RO Membrane : When the conductivity is under 35㎲/cm (RO-Pack Lamp "ON")

· Pre filtr va uglerod filtrini qanday almashtirish mumkin:



1) Suv berish klapanini yoping.

2) Bosimni pasaytirish uchun tizimni taxminan 10 ~ 20 soniya davomida ishlating

3) Tizim quvvatini o'chiring va quyidagi rasmdagi kabi bajaring.

4) Korpusni aylantirib ajrating.

5) Korpusdagi O-halqani tekshiring. Agar O-halqa yo'qolsa, suv oqishi mumkin.

6) Yangi filtr o'rnatilgandan so'ng, korpusga ulang.

7) Kran suvini oching, qurilmani ishga tushiring va sizib chiqish bor-yo'qligini tekshiring.

V. Filtrni almashtirish 2: RO membranasi

Quvvatni o'chiring va quyidagicha ishga tushiring.



※ Eslatma (Caution)

Agar almashtirilgandan keyin 1 soatdan so‘ng ko‘rsatkich 100 μS/cm dan yuqori bo‘lsa, “RO” va “Drain” o‘rtasidagi ulanishni tekshiring.

VI. Filtr almashinuvi 3: HP–Pack & 0.22 yakuniy filtr



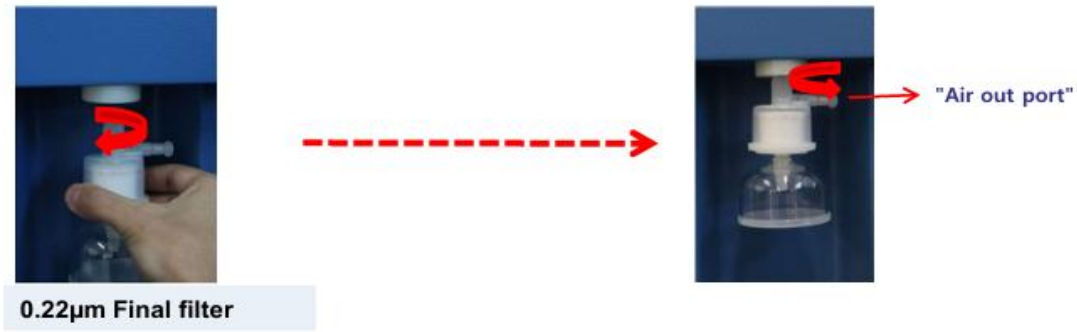
Tizim quvvatini o'chiring va quyidagi rasmdagi kabi bajaring.

·HP - Pack ni qanday almashtirish mumkin

Kalit (spanner) yordamida ulanishni (connector) ajrating.

Yangi HP-Pack ni korpusga joylashtiring, suv shlangini ulang va connectorni qayta birlashtiring.

·0.22μm yakuniy filtrni qanday almashtirish mumkin



1) 0,22 μm yakuniy filtrni burab chiqarib oling, yangisiga almashtiring va qayta burab mahkamlang.

2) Yakuniy filtrni almashtirganda, foydalanuvchi “Air out port” ni ochib havoni chiqarishi kerak, so‘ngra “Air out port” ni yopadi.

Ⅶ. O'rnatish



1. Level Sensor: Suv bakidan Level Sensor kabelini level sensor portiga ulang. Suv daraja sensorini bak qopqog‘iga mahkamlang.

2. Power Cable: Quvvat kabelini tizimning orqa qismidagi “Power” portiga ulang.

3. Drenaj (Drenaj shlangi): Ø6 suv shlangini “Drenaj” portidan chiqindi suvga ulang.

4. IN (kirish suvi shlangi): Ø10 mm suv shlangini suv manbasidan tizimning orqa qismidagi “IN” (kirish) portiga ulang.

5. RO (RO shlangi): Ø10 mm RO shlangini suv bakidan tizimning orqa qismidagi “RO” portiga ulang.

RO shlangi suv bakiga ulanganda, RO suv ta’minoti daraja sensori orqali avtomatik ravishda boshqariladi.

6. 0.22 Yakuniy filtr: Tizimning old qismidagi yuqoriga chiqish portini burab joylashtiring.

7. RO suv ishlab chiqarish:

1) Kran suvini oching.

2) “RO” tugmasini bosing.

3) “RESET” tugmasini bosing..

8. UP (ultra toza) suv ishlab chiqarish:

1) Krandan suv ta'minotini oching.

2) “UP” tugmasini bosing.

3) Ultra toza suv HP-Pack va 0,22 μm yakuniy filtr orqali ishlab chiqariladi.

9. Drenaj shlangi va RO Mahsulot shlangidagi suv oqimini tekshiring.

- Ishlab chiqarish va drenaj darajasi: 35% (Ishlab chiqarish), 65% (Drenaj)

• Diqqat: Atrof-muhit haroratini normal holatda saqlang.

Agar harorat 0℃ dan past bo‘lsa, suv tizimi jiddiy muzlashdan zarar ko‘rishi mumkin.

Ⅷ. Boshqaruv va tugmachalar paneli



Main Board

Boshqaruv
1. Tizim quvvatini yoqing (orqaga tugma) 2. Kirish suv klapanini oching (kran suvi). 3. “RO” suvini ishlab chiqarish uchun: “RO” va “Reset” tugmalarini bosing — RO suv ishlab chiqarila boshlaydi. Agar “RO” tugmasini yana bossangiz, qurilma to‘xtaydi. -RO diapazoni: 0 ~ 35㎲/sm (III turdagi) 4. “UP”(ultra toza) suvini ishlab chiqarish: “UP” tugmasini bosing. Agar “UP” tugmasini yana bossangiz, qurilma to‘xtaydi. -UP diapazoni: 10 ~ 18.3 ㏁-sm (I toifa)

Klaviatura paneli

MainPower	Tizim quvvat ko'rsatkichi
Tank	Suv baki daraja indikatori (Suv baki to‘lganda, chiroq yonadi)
Low Pressure	Past suv bosimi ko'rsatkichi (ixtiyoriy)
RO Pack	RO membranasini almashtirish indikatori (35㎲/sm dan yuqori)
HP Pack	HP-Pack almashtirish indikatori (10㏁-sm dan past)
RO	RO rejimi / RO to'xtatish tugmasi
Reset	“RO” suvni ishga tushirish tugmasi (“RO” tugmasi “ON” holatida bo‘lganda)
UP	UP suvni ishga tushirish / to‘xtatish tugmasi
Recycle	Ultra toza suv uchun aylanish (ixtiyoriy)

IX. Boshqaruv va tugmachalar paneli

Alomat	Tekshirish punkti	Yechim
RO ishlab chiqarish hajmi kamayishi	- RO qiymatini tekshiring (o'tkazuvchanlik): 0~35μs/sm - Drenaj suvini tekshiring - Kirish (feed) suvini va uning bosimini tekshiring - Nasos holatini tekshiring	- RO membranasini almashtiring (35μs/sm dan ortiq) - Drenaj klapanini sozlang (Mahsulot: 35%, Drenaj: 65%) - Kirish suvini ulang va bosimni tekshiring: Tavsiya etilgan bosim: 0,5 kg/cm² dan yuqori - Agar nasosda muammo bo'lsa, uni almashtiring
Suv baki to‘lib ketishi	- Level sensorni tekshiring (Level sensorni qo‘l bilan harakatlantirib ko‘ring)	- Sensorni almashtiring
Quvvatni o'chirish	- Quvvat ulanish portidagi himoya elementini (fuse) tekshiring - Quvvat manbasini tekshiring	- Himoya elementini (fuse) almashtiring - Muhandis bilan bog'lanish - Quvvat manbasini almashtiring
“RO to'plami”	- O'tkazuvchanlik qiymatini tekshiring	- RO membranasini almashtiring (35μs/sm dan ortiq)
“HP to‘plami”	- Qarshilik (rezistivlik) qiymatini tekshiring	- HP-Packni almashtiring (10 sm dan past)

Qisqa foydalanish yo‘riqnomasi

(Pure Power Ultra-Pure Water System)

1. Ishga tushirish

1-qadam: Kirish suvi klapanini (feed water valve) oching.

2-qadam: “RO” tugmasini bosing, so’ngra “Reset” tugmasini bosing.

Birlamchi tozalangan suv (1-bosqich) ishlab chiqariladi.

O'tkazuvchanlikni (Conductivity) tekshiring. (Normal diapazon: 0 – 35 µS/cm)

3-qadam: “UP” tugmasini bosing.

Ultra toza suv (3-bosqich) ishlab chiqariladi.

Qarshilikni (Resistivity) tekshiring. (Normal diapazon: 10.0 – 18.2 MΩ·cm)

2. Ishni to'xtatish (Shutdown)

⚠ **Har doim avval kirish suvi klapanini yoping, so'ngra “RO” va “UP” tugmalarini o'chiring (OFF).**

3. Asosiy kontroller indikatorlari (Main Controller Indicators)

Main Power (Asosiy quvvat): Quvvat berilganda chiroq yonadi.

Tank (Bak/Rezervuar): Asosiy suv baki to'lganda, tizim avtomatik to'xtaydi va chiroq yonadi.

RO Pack: RO filtrini almashtirish talab etilganda chiroq yonadi.

HP Pack: HP paketini almashtirish talab etilganda chiroq yonadi.

4. Filtrlarni almashtirish davri (Filter Replacement Cycle)

4-1. Pre-filtr: Oyiga bir marta almashtiring (yoki rangi jigarrang bo'lib qolsa).

4-2. Uglerod filtri (Carbon Filter): Har ikki pre-filtr almashtirilganda bir marta almashtiring.

4-3. RO membranasi: RO Pack chirog'i yonganda almashtiring.

4-4. HP paketi: HP Pack chirog'i yonganda almashtiring.

4-5. 0.22 µm yakuniy filtr (Final Filter): HP paketi almashtirilganda birga almashtiring.